Abstract

In this project, we developed a structured database and web application to explore and analyze Housing Development Board (HDB) resale flat prices in Singapore. Using data from data.gov.sg [2], specifically from January 2017 onward, we focused on tracking key housing trends such as average resale prices by flat type, town, storey range, and lease commencement year. We chose this date range because it aligns with the government's newer housing policies and ensures that our analysis is based on more recent and relevant transactions. By transforming the raw CSV dataset into a normalized relational database, we were able to efficiently run SQL queries that answer specific research questions. These insights are presented through a user-friendly web interface built with Node.js, Express [8], MySQL, and EJS. Our work highlights how database design and backend development can be used together to support deeper understanding of housing affordability and pricing patterns in Singapore.

Introduction

HDB flats play a central role in Singapore's housing landscape, making up the majority of homes for citizens. With rising interest in housing affordability and resale pricing trends, there is increasing value in exploring the resale flat market through structured data analysis. For this project, we chose to work with the HDB Resale Flat Prices dataset available on data.gov.sg [2], focusing on data starting from January 2017. This time frame was selected because it reflects more current market dynamics and housing policy changes, ensuring that our insights remain relevant to today's context.

We began by building a relational database to model the data, which includes details such as town, block, flat type, floor area, lease commencement year, storey range, and resale price. Using SQL and normalization techniques, we organized the dataset into efficient tables representing towns, blocks, flats, and transactions. To make the analysis interactive and user-friendly, we developed a web application that lets users explore various queries visually, such as identifying towns with the highest resale prices or understanding how lease year affects flat value.

This report documents our process from dataset selection and database modeling to implementation and query visualization. Through this project, we aim to show how structured data and web development tools can be combined to support meaningful housing data exploration in a local context.

Stage 1: Dataset Selection and Critique

This report explores the HDB Resale Flat Prices dataset, available on Singapore's data.gov.sg [1]. The dataset captures monthly transactions of resale flats, providing key details such as town, flat type, floor area, lease start date, and resale price. Its structured format makes it ideal for building a relational model and examining long-term housing trends in Singapore.

Dataset Source and Accessibility

The dataset is maintained by the Housing & Development Board (HDB) and hosted on the government's official open data portal, data.gov.sg [1]. It is published under the Singapore Open Data Licence v1.0 [2], which allows the data to be freely used, shared, and modified, making it accessible for both public and academic use.

- **Credibility**: The data comes from a trusted government source.
- **Accessibility**: It's downloadable in CSV format, and metadata is clearly provided.
- **License**: The licence is flexible and encourages educational use.

Quality and Detail

This dataset provides comprehensive information for each resale transaction. It includes:

- Town and block information
- Type of flat and size in square meters
- Storey range and lease commencement year
- Final resale price

These attributes offer rich possibilities for structured storage and querying using SQL.

- **Potential Limitation**: The dataset doesn't contain details like buyer or seller demographics, or whether the flat has been renovated which are factors that might also affect pricing.

- **Scope Note**: It's worth noting that this dataset only covers resale flats, not new sales or rental transactions.

Documentation and Use

The documentation provided is clear enough for users to understand each column, and the dataset has seen widespread use in government planning, academic research, and data visualizations [1].

- **Strengths**: Great for tracking price trends, comparing towns, and performing data analysis across time.
- **Improvement Area**: A bit more detail in the documentation, like how storey ranges are defined, could help users interpret the data more precisely.

Interrelation and Discoverability

The way the data is structured allows it to be easily connected to other datasets, such as:

- Town or region-level population data
- Income or employment statistics
- Public transport accessibility data

Future Potential: Linking this with other datasets could lead to deeper insights about how location and accessibility affect housing affordability.

Interest and Research Questions

Housing is a topic that affects almost everyone in Singapore, so it's no surprise that this dataset is both timely and relevant. It gives a data-backed way to explore how flat prices vary across time, locations, and flat types.

Research Questions

1. Which towns have the highest average resale price for 4-room flats?
2. How have resale prices changed over time across different flat types?
3. Is there a correlation between lease commencement year and resale value?
4. Do higher-floor flats consistently command higher prices?
5. Which flat types are most frequently transacted in each town?

These are the kinds of questions that benefit from being explored with a relational database, rather than a basic spreadsheet. A database lets us organize and query the data in more powerful and scalable ways.

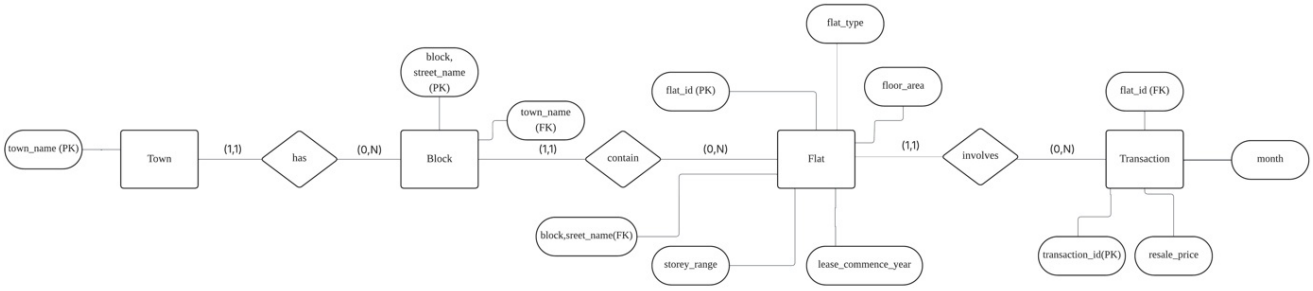
Stage 2: Model Your Data

This section explains the data modeling process used for the HDB resale flat price dataset. It begins with the design of an Entity Relationship diagram, which is later converted into a relational schema. This approach helps ensure that the structure of the database is both logical and efficient. The design follows best practices covered in the module materials and class discussions [1].

Entity-Relationship Diagram

The Entity Relationship diagram provides a clear overview of the key elements in the dataset, which include towns, blocks, flats, and transactions, along with the relationships between them. These entities and their attributes are based on the structure of the HDB resale flat dataset. To accurately reflect how they are connected, cardinality is used to represent one to many relationships across different layers of housing data, including locations, individual flats, and their corresponding resale records [2].

To better understand how this model works in the real world, consider the town of Bishan in Singapore. Bishan includes several HDB blocks, such as Block 101. Within this block, there may be multiple flats, like unit #12-123, which might have been sold once in 2015 and again in 2022. In this case, Bishan is the town, Block 101 is the block, unit #12-123 is the flat, and each sale is recorded as a separate transaction. This example clearly demonstrates the one-to-many structure of the data, where a town can include many blocks, a block can contain many flats, and a flat can be linked to multiple transactions over time.



ER Diagram Creation:

This Entity Relationship diagram outlines four main entities: Town, Block, Flat, and Transaction. Each one includes its own set of attributes and a designated primary key. The connections between these entities follow the standard notation taught in this module [1]. To clearly show how they relate, formal cardinality is used. For example, a Town may be linked to zero or many Blocks, while each Block is connected to exactly one Town. This structure helps maintain accuracy in how the data is organised and makes it easier to run consistent and meaningful queries across the database.

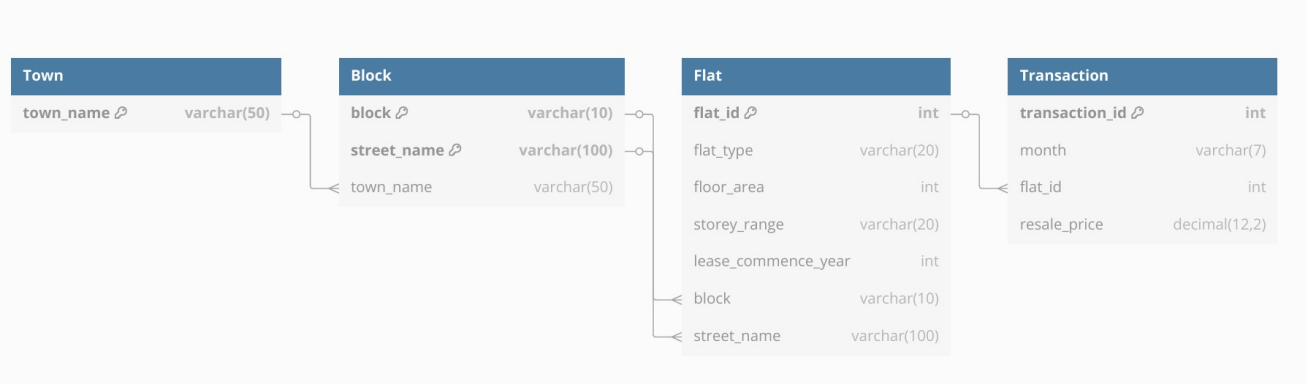
The diagram also reflects real-life structure and relationships within the HDB resale data. For instance, a Town can have many Blocks, but each Block belongs to one specific Town. Similarly, a Block can contain many Flats, but each Flat belongs to exactly one Block. Each Flat can be involved in multiple Transactions, representing its resale history over time, while each Transaction relates to one specific Flat. These one-to-many relationships are important for maintaining data integrity and supporting complex queries, such as tracking resale trends across different towns or identifying flats with multiple transactions.

Attributes and Relationships:

Attributes	Description
Town: town_name (PK)	The Town entity has a single attribute town_name as the primary key.
Block: block, street_name (PK), town_name (FK)	The Block entity uses a composite primary key consisting of block and street_name, with a foreign key town_name linking it to the Town entity.
Flat: flat_id (PK), flat_type, floor_area, storey_range, lease_commence_year, block, street_name (FK)	The Flat entity has flat_id as its primary key, and foreign keys block and street_name to link it to the Block entity. Other attributes describe the flat's details.
Transaction: transaction_id (PK), month, resale_price, flat_id (FK)	The Transaction entity tracks each resale transaction, with transaction_id as the primary key and flat_id as a foreign key linking back to the Flat entity.

Relational Schema

The Entity-Relationship diagram has been transformed into a relational schema that clearly shows how the data will be stored in a relational database. This schema lists the tables, their attributes, and how they are connected using primary and foreign keys. This design helps keep the data accurate, allows for efficient searches, and makes managing the HDB resale flat data straightforward.



Schema Translation:
The E/R diagram has been turned into four related tables: Town, Block, Flat, and Transaction. Each table has a unique ID to keep track of its records, and foreign keys link the tables together so everything stays connected and consistent.

Tables Description:

- **Town:** Uses the town's name (town_name) as its unique ID to store each town's name.
- **Block:** Has a combined ID made up of block and street_name. It also includes town_name to show which town each block belongs to.
- **Flat:** Has a unique ID called flat_id and keeps details like the flat type, floor area, storey range, and when the lease started. It also links to the block and street it belongs to using block and street_name.
- **Transaction:** Has a unique ID called transaction_id and records when the transaction happened (month), the resale price, and which flat was sold (flat_id).

Referential Integrity:
The foreign keys make sure everything is connected properly: blocks link to towns, flats link to blocks, and transactions link to flats. This helps keep the data accurate and makes it easy to ask complex questions that need information from multiple tables.

Database Normalization and Analysis

Normalization is an important step to organize the HDB resale flat price database so the data is clear and free of errors. By following

normalization rules, we reduce repeated data and avoid mistakes. This is especially important because the data has many connected parts like towns, blocks, flats, and resale transactions [4].

We checked our database design using the first three normal forms called 1NF, 2NF, and 3NF to make sure it stores data properly and allows easy searching. We also confirmed it follows Boyce-Codd Normal Form or BCNF to ensure the design is strong and reliable [5].

- First Normal Form means each table stores simple, separate pieces of data. For example, each transaction shows only one resale event and does not mix many values in one field [6].
- Second Normal Form means all information depends fully on the main key. Since each table has one main key, details like flat type or floor size depend only on that flat's ID [4].
- Third Normal Form means there are no hidden links between other data fields. For example, the lease start year or storey range depends only on the flat's ID and not on other fields [5].

Our design also follows BCNF which means all data depends on keys that uniquely identify each record. This helps avoid duplicate data and problems when updating information [4].

Choosing BCNF strikes a good balance by keeping data clean while still making searching easy. Since our data mainly tracks resale transactions without complex multiple values, going beyond BCNF was not necessary.

In summary, normalization helps make the database reliable by reducing repeated data like not storing the same town or block details many times. It also ensures updates such as price changes are applied correctly everywhere. This clear and tidy design makes it easier to study resale trends and housing patterns across towns and flats [7].

Functional Dependencies in the HDB Resale Flat Database

Functional dependencies explain how different pieces of data in the database relate to each other. Understanding these relationships helps keep the database organized and accurate.

Here are the main functional dependencies:

- The town's name uniquely identifies each town.
- The block and street name together uniquely identify each block, and each block belongs to one town.
- Each flat has a unique ID that determines all details about the flat, such as its type, floor area, storey range, lease start year, and the block and street it belongs to.
- Each transaction has a unique ID that identifies details like when the transaction happened, the resale price, and which flat was sold.

These dependencies show that all information depends fully on the main keys, without partial or hidden dependencies. This means the database follows the third normal form, which helps avoid mistakes and keeps the data consistent.

Why Functional Dependencies Matter for Normalization

These dependencies show that all non-key fields depend fully on their primary key, with no partial or hidden dependencies between non key fields. This confirms that the database design meets the requirements of the Third Normal Form (3NF), helping to ensure data consistency and avoid redundancy [5].

Stage 3. Create the database

Stage 3 involves the implementation of the database schema. This includes dropping any existing versions, creating normalized tables for the key entities, establishing referential integrity, and loading raw data into a staging table. These steps form the backbone for accurate and efficient data storage and retrieval. After building the database, we use SQL queries to dig into the data and find answers to key research questions. This helps us understand resale prices, how they change over time, and what factors affect them.

Drop and create the database

We will be using hdb_resale as my main database. In this stage, we set up the database that will store all the data for the project. First, we remove any existing database named hdb_resale to ensure a clean start. Then, we create a new database called hdb_resale and switch to it, so that all following operations will be done within this database.

In [ ]:
```
DROP DATABASE IF EXISTS hdb_resale;
CREATE DATABASE hdb_resale;
USE hdb_resale;
```

After that, we create a new database user named root with a password, giving this user permission to access and modify everything inside the hdb_resale database. This helps control who can work with the database and keeps things secure.

In [ ]:
```
DROP USER IF EXISTS 'root'@'%';
CREATE USER 'root'@'%' IDENTIFIED BY 'T0123667e-';

GRANT ALL ON hdb_resale.* TO 'root'@'%';
```

Create the normalized tables

To keep the resale flat data neat and easy to work with, we broke it down into several smaller tables. This helps avoid repeating the same info and makes sure everything stays accurate. We set up tables for:

- Towns: Just the unique town names.
- Blocks: Different blocks within each town, identified by block number and street name.
- Flats: Details about each flat, linked to its block.
- Transactions: Records of each resale, showing when it happened and for how much, connected to the right flat.

Below are the commands we used to create these tables and set up the relationships between them, so the data all fits together properly.

CREATE commands	Description
CREATE TABLE Town (town_name VARCHAR(50) PRIMARY KEY,);	This command creates the Town table, where each town is uniquely identified by its name, which acts as the primary key. It stores the names of towns.
CREATE TABLE Block (block VARCHAR(10), street_name VARCHAR(100), town_name VARCHAR(50), PRIMARY KEY (block, street_name), FOREIGN KEY (town_name) REFERENCES Town(town_name));	This creates the Block table, uniquely identified by the combination of block and street name. It stores block and street details and links each block to a town through a foreign key relationship.
CREATE TABLE Flat (flat_id INT AUTO_INCREMENT PRIMARY KEY, flat_type VARCHAR(20), floor_area INT, storey_range VARCHAR(20), lease_commence_year INT, block VARCHAR(10), street_name VARCHAR(100), FOREIGN KEY (block, street_name) REFERENCES Block(block, street_name));	This creates the Flat table, with a unique auto-incremented flat ID. It stores details such as flat type, floor area, storey range, and lease commencement year. Each flat is linked to a block and street through a foreign key.
CREATE TABLE Transaction (transaction_id INT AUTO_INCREMENT PRIMARY KEY, month VARCHAR(7), flat_id INT, resale_price DECIMAL(12,2), FOREIGN KEY (flat_id) REFERENCES Flat(flat_id));	This creates the Transaction table to record resale transactions. Each record has a unique transaction ID, sale month, resale price, and links to the corresponding flat.

Create the staging table to load raw CSV data

Before transferring data into the normalized tables, a temporary staging table named raw_resale_data was created to hold the raw information imported from the CSV file. This table uses appropriate data types to match the nature of each field, for example, the month is stored as a VARCHAR(7) to accommodate the "YYYY-MM" format, and resale_price uses a DECIMAL(12,2) type to precisely represent currency values. This staging area allows us to efficiently validate, clean, and preprocess the data before organizing it into the structured, normalized tables for more effective analysis and querying.

In [ ]:
```
CREATE TABLE raw_resale_data (
    month VARCHAR(7),
    town VARCHAR(50),
    flat_type VARCHAR(20),
    block VARCHAR(10),
    street_name VARCHAR(100),
    storey_range VARCHAR(20),
    floor_area INT,
    flat_model VARCHAR(50),
    lease_commence_year INT,
    resale_price DECIMAL(12,2)
);
```

Load CSV data into the staging table

Next, we loaded the resale flat data into a temporary staging table. This raw data comes from a CSV file containing resale transactions from January 2017 up to the present day. Using this staging table allows us to efficiently import and review the data before organizing it

into the normalized tables.

The command below shows how we imported the CSV file into the raw_resale_data staging table, specifying how the fields and lines are separated, and skipping the header row:

In [ ]:
```
## Coursera Lab version (use this one on Coursera platform)
LOAD DATA LOCAL INFILE '/home/coder/project/jolin database hdb mids/hdb_database/resale.csv'
INTO TABLE raw_resale_data
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;

## Local machine version (use this one on your own Windows PC)
LOAD DATA LOCAL INFILE 'C:\\Users\\jolin\\Downloads\\ResaleFlatPrices\\Resale flat prices based on registration
INTO TABLE raw_resale_data
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\r\n'
IGNORE 1 ROWS;
```

In [ ]:
```
SELECT * FROM raw_resale_data LIMIT 10;
```

```
mysql> SELECT * FROM raw_resale_data LIMIT 10;
+---------+------------+-----------+-------+--------------------+--------------+-----------+----------------+---------------------+--------------+
| month   | town       | flat_type | block | street_name        | storey_range | floor_area | flat_model     | lease_commence_year | resale_price |
+---------+------------+-----------+-------+--------------------+--------------+-----------+----------------+---------------------+--------------+
| 2017-01 | ANG MO KIO | 2 ROOM    | 406   | ANG MO KIO AVE 10  | 10 TO 12     |        44 | Improved       |                1979 |    232000.00 |
| 2017-01 | ANG MO KIO | 3 ROOM    | 108   | ANG MO KIO AVE 4   | 01 TO 03     |        67 | New Generation |                1978 |    250000.00 |
| 2017-01 | ANG MO KIO | 3 ROOM    | 602   | ANG MO KIO AVE 5   | 01 TO 03     |        67 | New Generation |                1980 |    262000.00 |
| 2017-01 | ANG MO KIO | 3 ROOM    | 465   | ANG MO KIO AVE 10  | 04 TO 06     |        68 | New Generation |                1980 |    265000.00 |
| 2017-01 | ANG MO KIO | 3 ROOM    | 601   | ANG MO KIO AVE 5   | 01 TO 03     |        67 | New Generation |                1980 |    265000.00 |
| 2017-01 | ANG MO KIO | 3 ROOM    | 150   | ANG MO KIO AVE 5   | 01 TO 03     |        68 | New Generation |                1981 |    275000.00 |
| 2017-01 | ANG MO KIO | 3 ROOM    | 447   | ANG MO KIO AVE 10  | 04 TO 06     |        68 | New Generation |                1979 |    280000.00 |
| 2017-01 | ANG MO KIO | 3 ROOM    | 218   | ANG MO KIO AVE 1   | 04 TO 06     |        67 | New Generation |                1976 |    285000.00 |
| 2017-01 | ANG MO KIO | 3 ROOM    | 447   | ANG MO KIO AVE 10  | 04 TO 06     |        68 | New Generation |                1979 |    285000.00 |
| 2017-01 | ANG MO KIO | 3 ROOM    | 571   | ANG MO KIO AVE 3   | 01 TO 03     |        67 | New Generation |                1979 |    285000.00 |
+---------+------------+-----------+-------+--------------------+--------------+-----------+----------------+---------------------+--------------+
```

Database Table Creation Commands and Descriptions

After creating the normalized tables, we populated them with data extracted from the staging table. This step organizes the raw resale flat data into structured tables by inserting unique records to avoid duplication.

Insert distinct towns into Town table

We insert all unique town names from the raw data into the Town table.

In [ ]:
```
INSERT IGNORE INTO Town (town_name)
SELECT DISTINCT town
FROM raw_resale_data;
```

Insert distinct blocks into Block table

Next, we insert distinct block and street name combinations along with their associated town into the Block table.

In [ ]:
```
INSERT IGNORE INTO Block (block, street_name, town_name)
SELECT DISTINCT block, street_name, town
FROM raw_resale_data;
```

Insert distinct flats into Flat table

We then populate the Flat table with unique flats, including details such as flat type, floor area, storey range, lease commencement year, and location (block and street).

In [ ]:
```
INSERT IGNORE INTO Flat (flat_type, floor_area, storey_range, lease_commence_year, block, street_name)
SELECT DISTINCT flat_type, floor_area, storey_range, lease_commence_year, block, street_name
FROM raw_resale_data;
```

Insert transactions into Transaction table

Finally, the resale transactions are inserted into the Transaction table. We link each transaction to the corresponding flat using the flat ID. This involves joining the staging table with the Flat table based on multiple matching attributes.

In [ ]:
```
INSERT IGNORE INTO Transaction (month, flat_id, resale_price)
SELECT
    month,
    f.flat_id,
    resale_price
```

```
FROM raw_resale_data r
JOIN Flat f ON r.flat_type = f.flat_type
    AND r.floor_area = f.floor_area
    AND r.storey_range = f.storey_range
    AND r.lease_commence_year = f.lease_commence_year
    AND r.block = f.block
    AND r.street_name = f.street_name;
```

Verifying data in the normalized Tables

After inserting the data into the normalized tables, we verified that the data was correctly populated by selecting a sample of records from each table. This step helps confirm the integrity of the data before proceeding with any analysis or further processing.

In [ ]:
```
SELECT * FROM Town LIMIT 5;
SELECT * FROM Block LIMIT 5;
SELECT * FROM Flat LIMIT 5;
SELECT * FROM Transaction LIMIT 5;
```

```
mysql> SELECT * FROM Town LIMIT 5;
+-------------+
| town_name   |
+-------------+
| ANG MO KIO  |
| BEDOK       |
| BISHAN      |
| BUKIT BATOK |
| BUKIT MERAH |
+-------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Block LIMIT 5;
+-------+------------------+------------+
| block | street_name      | town_name  |
+-------+------------------+------------+
| 101   | ANG MO KIO AVE 3 | ANG MO KIO |
| 102   | ANG MO KIO AVE 3 | ANG MO KIO |
| 103   | ANG MO KIO AVE 3 | ANG MO KIO |
| 104A  | ANG MO KIO ST 11 | ANG MO KIO |
| 104B  | ANG MO KIO ST 11 | ANG MO KIO |
+-------+------------------+------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Flat LIMIT 5;
+---------+-----------+------------+--------------+---------------------+-------+------------------+
| flat_id | flat_type | floor_area | storey_range | lease_commence_year | block | street_name      |
+---------+-----------+------------+--------------+---------------------+-------+------------------+
|       1 | 2 ROOM    |         44 | 10 TO 12     |                1979 | 406   | ANG MO KIO AVE 10 |
|       2 | 3 ROOM    |         67 | 01 TO 03     |                1978 | 108   | ANG MO KIO AVE 4  |
|       3 | 3 ROOM    |         67 | 01 TO 03     |                1980 | 602   | ANG MO KIO AVE 5  |
|       4 | 3 ROOM    |         68 | 04 TO 06     |                1980 | 465   | ANG MO KIO AVE 10 |
|       5 | 3 ROOM    |         67 | 01 TO 03     |                1980 | 601   | ANG MO KIO AVE 5  |
+---------+-----------+------------+--------------+---------------------+-------+------------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Transaction LIMIT 5;
+----------------+---------+---------+--------------+
| transaction_id | month   | flat_id | resale_price |
+----------------+---------+---------+--------------+
|              1 | 2017-01 |       1 |    232000.00 |
|              2 | 2017-01 |       2 |    250000.00 |
|              3 | 2017-01 |       3 |    262000.00 |
|              4 | 2017-01 |       4 |    265000.00 |
|              5 | 2017-01 |       5 |    265000.00 |
+----------------+---------+---------+--------------+
```

Reflection on Database Implementation

My database accurately represents the original HDB resale data. By organizing the data into clear tables for towns, blocks, flats, and transactions, I kept things neat and avoided repeating information. Using a staging table helped me import the raw data smoothly before breaking it down into these tables.

I made sure to match the data types closely to the real data, like storing prices as decimals and months as text in the year-month format, so nothing important gets lost or changed. Although I ran into some messy data and duplicates, the final setup captures the key details well and keeps the relationships clear.

SQL Queries to Answer Research Questions

In this part, we use the implemented database to answer specific research questions related to HDB resale trends. We write and run SQL queries that analyze the data to reveal insights such as which towns have the highest average resale prices for certain flat types, how resale prices change over time, and other patterns related to lease commencement, floor levels, and transaction frequencies. By doing this, we aim to better understand the resale market dynamics using the structured data we prepared.

For most of the questions, we focus on the average resale price rather than individual sale prices. To ensure clarity and precision, the resale prices are formatted using FORMAT(ROUND(AVG(tr.resale_price), 2), 2) AS avg_resale_price. This means the average prices are rounded to two decimal places and displayed with consistent formatting.

Using the average resale price helps smooth out any extreme values or one-off sales, providing a clearer and more reliable picture of general market trends and typical pricing. This approach also makes it easier to compare resale prices across different towns, flat types, and time periods.

1. Top 10 Towns with Highest Average Resale Price for 4-Room Flats

This SQL query calculates the average resale price of 4-room flats for each town. It joins the Transaction, Flat, Block, and Town tables to link sales data with flat details and their locations. The query filters to include only flats labeled as "4 ROOM." Then, it groups the data by town name and calculates the average resale price for each town. The results are sorted from highest to lowest average price, showing the top ten towns.

In [ ]:
```
SELECT
    t.town_name,
    FORMAT(ROUND(AVG(tr.resale_price), 2), 2) AS avg_resale_price
FROM
    Transaction tr
JOIN Flat f ON tr.flat_id = f.flat_id
JOIN Block b ON f.block = b.block AND f.street_name = b.street_name
JOIN Town t ON b.town_name = t.town_name
WHERE
    f.flat_type = '4 ROOM'
GROUP BY
    t.town_name
ORDER BY
    AVG(tr.resale_price) DESC
LIMIT 10;
```

town_name	avg_resale_price
CENTRAL AREA	832,478.15
QUEENSTOWN	782,115.33
BUKIT MERAH	730,604.69
KALLANG/WHAMPOA	692,294.67
TOA PAYOH	671,109.81
BUKIT TIMAH	668,209.07
CLEMENTI	631,380.00
BISHAN	617,328.05
GEYLANG	605,969.09
ANG MO KIO	547,711.41

2. Resale Price Trends Over Time by Flat Type

This query calculates the average resale prices for each flat type over the years. It groups the data by flat type and year, then finds the average resale price for each group. The year is taken from the month column and the prices are rounded to two decimal places to make them easier to read. The results are sorted by year and flat type so you can see how resale prices for different flat types, like one-room or executive flats, have changed from 2017 up to 2025.

In [ ]:
```
SELECT
  f.flat_type,
  CAST(LEFT(tr.month, 4) AS UNSIGNED) AS year,
  FORMAT(ROUND(AVG(tr.resale_price), 2), 2) AS avg_resale_price
FROM
  Transaction tr
JOIN
  Flat f ON tr.flat_id = f.flat_id
GROUP BY
  f.flat_type,
  year
ORDER BY
  year ASC,
  f.flat_type ASC;
;
```

flat_type	year	avg_resale_price
1 ROOM	2017	200,888.89
2 ROOM	2017	240,354.22
3 ROOM	2017	317,834.24
4 ROOM	2017	437,120.00
5 ROOM	2017	532,276.72
EXECUTIVE	2017	627,211.27
MULTI-GENERATION	2017	783,750.00
1 ROOM	2018	184,111.11
2 ROOM	2018	235,763.40
3 ROOM	2018	306,477.98
4 ROOM	2018	431,752.94
5 ROOM	2018	527,634.51
EXECUTIVE	2018	630,780.13
MULTI-GENERATION	2018	801,392.53
1 ROOM	2019	175,842.55
2 ROOM	2019	228,534.13
3 ROOM	2019	299,457.24
4 ROOM	2019	429,748.88
5 ROOM	2019	526,811.63
EXECUTIVE	2019	617,561.23
MULTI-GENERATION	2019	819,190.29
1 ROOM	2020	180,000.00
2 ROOM	2020	233,989.09
3 ROOM	2020	312,708.74
4 ROOM	2020	448,608.28
5 ROOM	2020	541,457.42
EXECUTIVE	2020	635,394.96
MULTI-GENERATION	2020	773,125.00
1 ROOM	2021	197,222.22
2 ROOM	2021	267,665.61
3 ROOM	2021	352,960.74
4 ROOM	2021	505,094.85
5 ROOM	2021	603,989.60
EXECUTIVE	2021	705,559.08
MULTI-GENERATION	2021	820,777.78
1 ROOM	2022	230,888.80
2 ROOM	2022	297,564.57
3 ROOM	2022	388,974.34
4 ROOM	2022	549,087.57
5 ROOM	2022	654,253.27
EXECUTIVE	2022	782,007.03
MULTI-GENERATION	2022	874,949.00
1 ROOM	2023	243,000.00
2 ROOM	2023	321,432.08
3 ROOM	2023	411,871.74
4 ROOM	2023	584,049.95
5 ROOM	2023	685,338.09
EXECUTIVE	2023	829,310.25
MULTI-GENERATION	2023	996,666.67
1 ROOM	2024	253,363.64
2 ROOM	2024	340,887.42
3 ROOM	2024	439,928.34
4 ROOM	2024	627,383.77
5 ROOM	2024	728,665.81
EXECUTIVE	2024	874,650.59
MULTI-GENERATION	2024	1,049,000.00
1 ROOM	2025	265,000.00
2 ROOM	2025	359,952.53
3 ROOM	2025	468,941.41
4 ROOM	2025	668,815.63
5 ROOM	2025	774,399.06
EXECUTIVE	2025	918,796.73
MULTI-GENERATION	2025	1,167,962.67

3. Correlation Between Lease Commencement Year and Resale Value

This SQL query finds the average resale price of flats based on their lease commencement year. It joins the Flat table with the Transaction table to link resale prices to each flat's lease start year. The results are grouped by the lease commencement year and sorted in ascending order.

This helps us see how the age of a flat's lease correlates with its resale value. Generally, flats with newer leases tend to have higher average resale prices, reflecting the value impact of lease length.

In [ ]:
```
SELECT
    f.lease_commence_year,
    FORMAT(AVG(tr.resale_price), 2) AS avg_resale_price
FROM
    Flat f
JOIN
    Transaction tr ON f.flat_id = tr.flat_id
GROUP BY
    f.lease_commence_year
ORDER BY
    f.lease_commence_year;
```

lease_commence_year	avg_resale_price
1966	254,277.78
1967	282,007.08
1968	325,296.69
1969	291,288.43
1970	311,323.05
1971	287,957.77
1972	380,468.66
1973	376,093.48
1974	395,577.60
1975	443,964.43
1976	395,990.27
1977	419,585.45
1978	409,998.77
1979	400,994.39
1980	408,310.85
1981	398,559.23
1982	425,339.21
1983	421,744.98
1984	471,959.72
1985	451,006.53
1986	480,173.11
1987	487,186.26
1988	493,717.20
1989	533,314.95
1990	510,740.75
1991	546,377.87
1992	634,457.51
1993	559,994.93
1994	602,099.68
1995	607,707.10
1996	563,492.15
1997	540,390.28
1998	543,283.91
1999	532,445.06
2000	525,623.18
2001	520,900.96
2002	522,034.32
2003	529,305.87
2004	492,335.36
2005	606,421.09
2006	694,276.52
2007	526,596.65
2008	617,670.24
2009	656,790.87
2010	541,363.08
2011	826,883.13
2012	572,803.45
2013	565,349.00
2014	523,195.54
2015	497,932.07
2016	595,352.24
2017	631,267.36
2018	627,345.49
2019	627,727.95
2020	726,475.37
2021	675,035.95

What it does: This query calculates the average resale price of flats grouped by the year their lease started. It helps analyze if older or newer leases affect resale prices, showing trends based on lease commencement year.

4. Do Higher Floors Command Higher Prices?

This SQL query investigates whether flats on higher floors generally have higher resale prices. It calculates the average resale price for flats grouped by their storey range, such as "01 TO 03", "04 TO 06", and so forth.

By joining the Flat and Transaction tables, the query links resale prices to their corresponding storey ranges. The results are ordered by storey range to reveal how average resale prices vary as you move from lower to higher floors.

From the data, you can observe a clear trend: average resale prices tend to increase as the storey range rises, with prices starting around $466,216 for floors 1 to 3 and reaching over$1,200,000 for floors 49 to 51. This suggests that flats on higher floors command a significant price premium, likely due to factors such as better views, reduced noise, and increased privacy.

However, it is worth noting that the price increase is not strictly linear, and the highest floors might also be influenced by other considerations such as building design or accessibility. Overall, the trend indicates that higher floors are more valued in the resale market.

In [ ]:
```
SELECT
    f.storey_range,
    FORMAT(AVG(tr.resale_price), 2) AS avg_resale_price
FROM
    Flat f
JOIN
    Transaction tr ON f.flat_id = tr.flat_id
GROUP BY
    f.storey_range
ORDER BY
    f.storey_range;
```

storey_range	avg_resale_price
01 TO 03	466,216.65
04 TO 06	485,844.07
07 TO 09	499,550.07
10 TO 12	512,971.48
13 TO 15	552,750.51
16 TO 18	594,373.21
19 TO 21	661,485.54
22 TO 24	688,558.60
25 TO 27	742,356.23
28 TO 30	829,395.79
31 TO 33	878,186.91
34 TO 36	885,618.87
37 TO 39	900,901.89
40 TO 42	970,746.48
43 TO 45	1,076,483.13
46 TO 48	1,131,352.29
49 TO 51	1,208,212.61

5. Most Frequently Transacted Flat Types in Each Town

This query finds the most frequently transacted flat types in each town. It joins the Town, Block, Flat, and Transaction tables to count how many resale transactions occurred for each flat type in each town. The results are grouped by town and flat type, and ordered by transaction count in descending order.

You can see, for example, that in ANG MO KIO, 3-room flats had the highest number of transactions, followed by 4-room flats and then 5-room flats. This helps identify which flat types are most popular in terms of resale activity across different towns.

In [ ]:
```
SELECT
    t.town_name,
    f.flat_type,
    COUNT(*) AS transaction_count
FROM
    Town t
JOIN
    Block b ON t.town_name = b.town_name
JOIN
    Flat f ON b.block = f.block AND b.street_name = f.street_name
JOIN
    Transaction tr ON f.flat_id = tr.flat_id
GROUP BY
    t.town_name,
    f.flat_type
ORDER BY
    t.town_name,
    transaction_count DESC;
```

town_name	flat_type	transaction_count
ANG MO KIO	3 ROOM	4546
ANG MO KIO	4 ROOM	2438
ANG MO KIO	5 ROOM	1271
ANG MO KIO	2 ROOM	156
ANG MO KIO	EXECUTIVE	125
BEDOK	3 ROOM	4468
BEDOK	4 ROOM	3723
BEDOK	5 ROOM	2007
BEDOK	EXECUTIVE	542
BEDOK	2 ROOM	145
BISHAN	4 ROOM	1776
BISHAN	5 ROOM	1105
BISHAN	3 ROOM	405
BISHAN	EXECUTIVE	377
BISHAN	MULTI-GENERATION	13
BUKIT BATOK	4 ROOM	3639
BUKIT BATOK	3 ROOM	2570
BUKIT BATOK	5 ROOM	1433
BUKIT BATOK	EXECUTIVE	667
BUKIT BATOK	2 ROOM	228
BUKIT MERAH	4 ROOM	3103
BUKIT MERAH	3 ROOM	2841
BUKIT MERAH	5 ROOM	1718
BUKIT MERAH	2 ROOM	243
BUKIT MERAH	1 ROOM	77
BUKIT MERAH	EXECUTIVE	3
BUKIT PANJANG	4 ROOM	3469
BUKIT PANJANG	5 ROOM	2224
BUKIT PANJANG	3 ROOM	902
BUKIT PANJANG	EXECUTIVE	774
BUKIT PANJANG	2 ROOM	133
BUKIT TIMAH	4 ROOM	193
BUKIT TIMAH	5 ROOM	120
BUKIT TIMAH	EXECUTIVE	105
BUKIT TIMAH	3 ROOM	94
CENTRAL AREA	3 ROOM	679
CENTRAL AREA	4 ROOM	659
CENTRAL AREA	5 ROOM	238
CENTRAL AREA	2 ROOM	65
CENTRAL AREA	EXECUTIVE	2
CHOA CHU KANG	4 ROOM	4597
CHOA CHU KANG	5 ROOM	3211
CHOA CHU KANG	EXECUTIVE	953
CHOA CHU KANG	3 ROOM	617
CHOA CHU KANG	2 ROOM	86

CLEMENTI	3 ROOM	2135
CLEMENTI	4 ROOM	1699
CLEMENTI	5 ROOM	591
CLEMENTI	EXECUTIVE	151
CLEMENTI	2 ROOM	43
GEYLANG	3 ROOM	2313
GEYLANG	4 ROOM	1840
GEYLANG	5 ROOM	607
GEYLANG	EXECUTIVE	199
GEYLANG	2 ROOM	164
HOUGANG	4 ROOM	4728
HOUGANG	5 ROOM	2287
HOUGANG	3 ROOM	2206
HOUGANG	EXECUTIVE	1097
HOUGANG	2 ROOM	167
JURONG EAST	3 ROOM	1468
JURONG EAST	4 ROOM	1274
JURONG EAST	5 ROOM	1013
JURONG EAST	EXECUTIVE	423
JURONG EAST	2 ROOM	81
JURONG WEST	4 ROOM	5146
JURONG WEST	5 ROOM	4339
JURONG WEST	3 ROOM	2659
JURONG WEST	EXECUTIVE	1457
JURONG WEST	2 ROOM	216
KALLANG/WHAMPOA	3 ROOM	2735
KALLANG/WHAMPOA	4 ROOM	2347
KALLANG/WHAMPOA	5 ROOM	1051
KALLANG/WHAMPOA	EXECUTIVE	133
KALLANG/WHAMPOA	2 ROOM	94
MARINE PARADE	3 ROOM	625
MARINE PARADE	5 ROOM	336
MARINE PARADE	4 ROOM	313
MARINE PARADE	2 ROOM	7
PASIR RIS	4 ROOM	2358
PASIR RIS	5 ROOM	1906
PASIR RIS	EXECUTIVE	1552
PASIR RIS	3 ROOM	155
PASIR RIS	2 ROOM	50
PUNGGOL	4 ROOM	8106
PUNGGOL	5 ROOM	4860
PUNGGOL	3 ROOM	1603
PUNGGOL	2 ROOM	420
PUNGGOL	EXECUTIVE	286
QUEENSTOWN	3 ROOM	2486
QUEENSTOWN	4 ROOM	2256
QUEENSTOWN	5 ROOM	743
QUEENSTOWN	2 ROOM	151
QUEENSTOWN	EXECUTIVE	56
SEMBAWANG	4 ROOM	2836
SEMBAWANG	5 ROOM	1844
SEMBAWANG	EXECUTIVE	663
SEMBAWANG	3 ROOM	425
SEMBAWANG	2 ROOM	348
SENGKANG	4 ROOM	8562
SENGKANG	5 ROOM	5846
SENGKANG	3 ROOM	1225
SENGKANG	EXECUTIVE	1082
SENGKANG	2 ROOM	422
SERANGOON	4 ROOM	1635
SERANGOON	3 ROOM	813
SERANGOON	5 ROOM	680
SERANGOON	EXECUTIVE	577
SERANGOON	2 ROOM	18
TAMPINES	4 ROOM	5803
TAMPINES	5 ROOM	3801
TAMPINES	3 ROOM	2971
TAMPINES	EXECUTIVE	1454
TAMPINES	2 ROOM	96
TAMPINES	MULTI-GENERATION	26
TOA PAYOH	3 ROOM	2907
TOA PAYOH	4 ROOM	2118
TOA PAYOH	5 ROOM	1217
TOA PAYOH	2 ROOM	199
TOA PAYOH	EXECUTIVE	168
WOODLANDS	4 ROOM	6653
WOODLANDS	5 ROOM	4493
WOODLANDS	3 ROOM	1831
WOODLANDS	EXECUTIVE	1614
WOODLANDS	2 ROOM	212
YISHUN	4 ROOM	6910
YISHUN	3 ROOM	3880
YISHUN	5 ROOM	2402
YISHUN	EXECUTIVE	678
YISHUN	2 ROOM	305
YISHUN	MULTI-GENERATION	44

Stage 4: Making the website

Web Application Setup Using Node.js and Express

To help answer our research questions, we built a web application using Node.js and the Express framework. This app connects to our MySQL database (hdb_resale) using a connection pool, which improves performance by handling multiple database requests at the same time. The server runs on port 3001 and uses EJS templates to generate dynamic pages based on the data retrieved from the database [8].

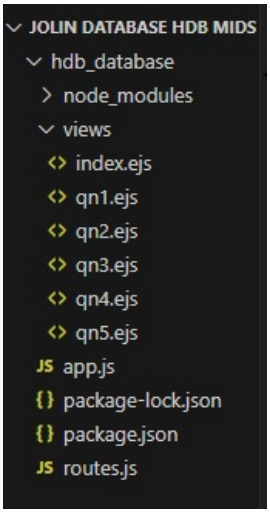
Database Access Configuration

To make this connection possible, we created a MySQL user named 'root' with the password 'T0123667e-', and gave it full access to the hdb_resale database using the GRANT ALL command. This ensures that our backend has the necessary permissions to run all the required queries smoothly and securely [9].

```
const express = require('express');
const path = require('path');
const mysql = require('mysql2');
const createRoutes = require('./routes');

const app = express();
const port = 3001;

app.set('view engine', 'ejs');
app.set('views', path.join(__dirname, 'views'));
app.use(express.static(path.join(__dirname, 'public')));

const pool = mysql.createPool({
  host: 'localhost',
  user: 'root',
  password: 'T0123667e-',
  database: 'hdb_resale',
  waitForConnections: true,
  connectionLimit: 10,
  queueLimit: 0,
});

// Use all GET routes from the external file
app.use('/', createRoutes(pool));

// Home page route
app.get('/', (req, res) => {
  res.render('index', { title: 'HDB Resale Data Reports' });
});

app.listen(port, () => {
  console.log(`Server running at http://localhost:${port}`);
});
```

Modular Project Structure

For better organization and easier maintenance, we structured our code into separate parts. The main application logic is handled in app.js, while all the route definitions are kept in a separate file (routes.js). This file is responsible for handling each query and sending the results to the correct EJS template. We also organized our front-end files inside a views/ folder, where index.ejs serves as the homepage and qn1.ejs to qn5.ejs each display the results for different queries.



Data Query Results

When users first access the web application, they are are presented with a clean and intuitive homepage.This page acts as a central menu where users can choose from five different data reports related to HDB resale prices.

The homepage is rendered through the following route in app.js:

```
// Home page route
app.get('/', (req, res) => {
  res.render('index', { title: 'HDB Resale Data Reports' });
});
```

Below is the final layout of the homepage as it appears to users when the application runs.

We start with our first query, which focuses on identifying the top 10 towns in Singapore with the highest average resale prices for 4-room HDB flats. This begins with the /top-towns route defined in our backend. Inside that route, we wrote a SQL query that brings together data from several related tables: Transaction, Flat, Block, and Town.

The query looks for flats specifically labeled as "4 ROOM" and calculates the average resale price for each town. It then sorts the towns from highest to lowest based on those average prices and limits the output to just the top 10 results.

```
module.exports = (pool) => {
  // Q1 - Top 10 Towns by Avg Resale Price
  router.get('/top-towns', (req, res) => {
    const sql = `
      SELECT t.town_name, FORMAT(ROUND(AVG(tr.resale_price), 2), 2) AS avg_resale_price
      FROM Transaction tr
      JOIN Flat f ON tr.flat_id = f.flat_id
      JOIN Block b ON f.block = b.block AND f.street_name = b.street_name
      JOIN Town t ON b.town_name = t.town_name
      WHERE f.flat_type = '4 ROOM'
      GROUP BY t.town_name
      ORDER BY AVG(tr.resale_price) DESC
      LIMIT 10;
    `;
    pool.query(sql, (err, results) => {
      if (err) return res.status(500).send('Database query error');
      res.render('qn1', { title: 'Top 10 Towns by Avg Resale Price', data: results });
    });
  });
```

Once the data is retrieved, it's passed into the qn1.ejs view file, where it's displayed neatly in a table. On the page, users can clearly see each town's name along with its corresponding average resale price.

Top 10 Towns by Avg Resale Price

Town Name	Average Resale Price ($)
CENTRAL AREA	832,478.15
QUEENSTOWN	782,115.33
BUKIT MERAH	730,604.69
KALLANG/WHAMPOA	692,294.67
TOA PAYOH	671,109.81
BUKIT TIMAH	668,209.07
CLEMENTI	631,380.00
BISHAN	617,328.05
GEYLANG	605,969.09
ANG MO KIO	547,711.41

Back to Home

The next query looks at how average resale prices change across different flat types and years. In the backend, we created a route called /avg-price-by-year. Inside that route, we wrote a SQL query that groups the data by both flat type and year. To extract the year from the month column, we used the LEFT function to grab the first four characters and cast it as an integer. The query then calculates the average resale price for each flat type in each year and sorts the results in ascending order.

```
// Q2 - Average Price by Flat Type and Year
router.get('/avg-price-by-year', (req, res) => {
  const sql = `
    SELECT f.flat_type, CAST(LEFT(tr.month, 4) AS UNSIGNED) AS year,
           FORMAT(ROUND(AVG(tr.resale_price), 2), 2) AS avg_resale_price
    FROM Transaction tr
    JOIN Flat f ON tr.flat_id = f.flat_id
    GROUP BY f.flat_type, year
    ORDER BY year ASC, f.flat_type ASC;
  `;
  pool.query(sql, (err, results) => {
    if (err) return res.status(500).send('Database query error');
    res.render('qn2', { title: 'Average Resale Price by Flat Type & Year', data: results });
  });
});
```

Once the results are retrieved, they're passed to the qn2.ejs template, where the data is displayed in a table. At the top of the page, users can choose a specific year using the filter buttons. When a year is selected, the page shows all the flat types and their corresponding average resale prices for that year. This makes it easy for users to compare pricing trends across different flat types and observe how resale values have shifted over time.

Average Resale Price by Flat Type & Year

2017 2018 2019 2020 2021 2022 2023 2024 2025

Flat Type	Year	Average Resale Price ($)
1 ROOM	2021	197,222.22
2 ROOM	2021	267,665.61
3 ROOM	2021	352,960.74
4 ROOM	2021	505,094.85
5 ROOM	2021	603,989.60
EXECUTIVE	2021	705,559.08
MULTI-GENERATION	2021	820,777.78

Back to Home

This part of the application focuses on showing the average resale prices based on the lease commencement year of HDB flats. In the /lease-year-avg-price route, we wrote a SQL query that groups the data by each flat's lease commencement year. It calculates the average resale price for each year using the AVG() function, and formats the results to two decimal places for readability.

The data is pulled from the Flat and Transaction tables, joined by the flat_id, and sorted by lease year in ascending order. This helps users clearly see how resale prices vary depending on how old the flat is, starting from as early as the 1960s.

```
// Q3 - Average Price by Lease Year
router.get('/lease-year-avg-price', (req, res) => {
  const sql = `
    SELECT f.lease_commence_year, FORMAT(AVG(tr.resale_price), 2) AS avg_resale_price
    FROM Flat f JOIN Transaction tr ON f.flat_id = tr.flat_id
    GROUP BY f.lease_commence_year
    ORDER BY f.lease_commence_year;
  `;
  pool.query(sql, (err, results) => {
    if (err) return res.status(500).send('Database query error');
    res.render('qn3', { title: 'Average Resale Price by Lease Commence Year', data: results });
  });
});
```

The query results are sent to the qn3.ejs file, where the front end displays the data in a clean two-column table: one for the lease year and one for the average resale price. This view gives users an easy way to observe pricing trends over time, which can be useful for understanding how the age of a flat influences its value in the resale market.

Average Resale Price by Lease Commence Year

Lease Commence Year	Average Resale Price ($)
1966	254,277.78
1967	282,007.08
1968	325,296.69
1969	291,288.43
1970	311,323.05
1971	287,957.77
1972	380,468.66
1973	376,093.48
1974	395,577.60
1975	443,964.43
1976	395,990.27
1977	419,585.45
1978	409,998.77
1979	400,994.39
1980	408,310.85

This section of the web application explores how average resale prices differ across various storey ranges. To do this, we created the /avg-price-by-storey route and wrote a SQL query that groups the resale data by storey range. It calculates the average resale price for each group using the AVG() function and formats the result to two decimal places for clarity.

The query joins the Flat and Transaction tables through the flat_id, and then groups all flats based on their storey_range. Finally, the results are sorted by the storey range to ensure that the values appear in a logical, ascending order.

```
// Q4 - Average Price by Storey Range
router.get('/avg-price-by-storey', (req, res) => {
  const sql = `
    SELECT f.storey_range, FORMAT(AVG(tr.resale_price), 2) AS avg_resale_price
    FROM Flat f JOIN Transaction tr ON f.flat_id = tr.flat_id
    GROUP BY f.storey_range
    ORDER BY f.storey_range;
  `;
  pool.query(sql, (err, results) => {
    if (err) return res.status(500).send('Database query error');
    res.render('qn4', { title: 'Average Resale Price by Storey Range', data: results });
  });
});
```

Once the data is retrieved from the database, it's passed to the qn4.ejs file. On the front end, the table clearly shows each storey range alongside its corresponding average resale price. This makes it easy for users to spot trends, such as how flats located on higher floors tend to have higher average resale values.

Average Resale Price by Storey Range

Storey Range	Average Resale Price ($)
01 TO 03	466,216.65
04 TO 06	485,844.07
07 TO 09	499,550.07
10 TO 12	512,971.48
13 TO 15	552,750.51
16 TO 18	594,373.21
19 TO 21	661,485.54
22 TO 24	688,558.60
25 TO 27	742,356.23
28 TO 30	829,395.79
31 TO 33	878,186.91
34 TO 36	885,618.87
37 TO 39	900,901.89
40 TO 42	970,746.48
43 TO 45	1,076,483.13
46 TO 48	1,131,352.29
49 TO 51	1,208,212.61

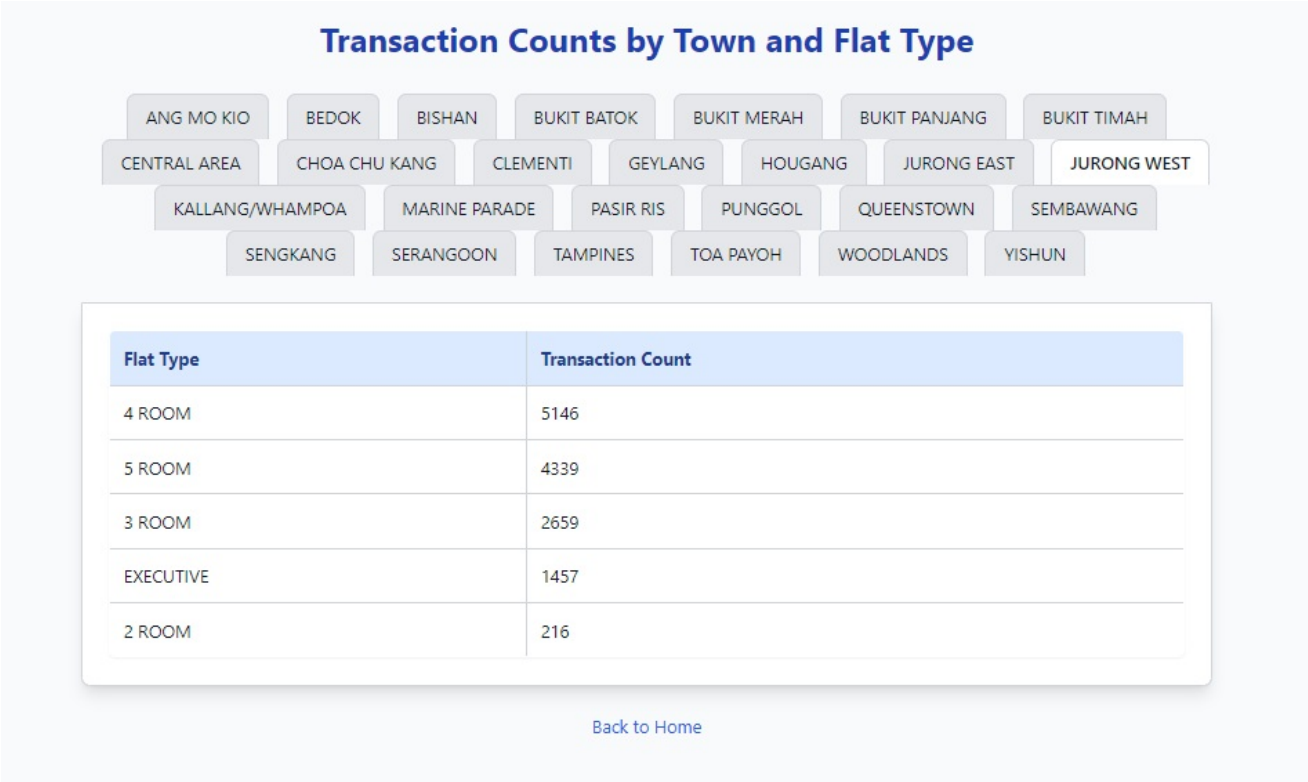
Back to Home

This part of the application helps users explore how many resale transactions took place in each town, grouped by flat type. In the backend, we set up a route called /transactions-by-town. Inside this route, we created a SQL query that pulls together data from several tables: Town, Block, Flat, and Transaction.

The query connects these tables using their related fields and groups the results by both town name and flat type. It then counts the number of transactions in each group using the COUNT(*) function. The results are sorted so that towns with more resale activity appear first.

```
  // Q5 - Transaction Counts by Town and Flat Type
  router.get('/transactions-by-town', (req, res) => {
    const sql = `
      SELECT t.town_name, f.flat_type, COUNT(*) AS transaction_count
      FROM Town t
      JOIN Block b ON t.town_name = b.town_name
      JOIN Flat f ON b.block = f.block AND b.street_name = f.street_name
      JOIN Transaction tr ON f.flat_id = tr.flat_id
      GROUP BY t.town_name, f.flat_type
      ORDER BY t.town_name, transaction_count DESC;
    `;
    pool.query(sql, (err, results) => {
      if (err) return res.status(500).send('Database query error');
      res.render('qn5', { title: 'Transaction Counts by Town and Flat Type', data: results });
    });
  });

  return router;
};
```

Once the data is ready, it is passed into the qn5.ejs file, where it is shown in a clean and easy-to-read table. At the top of the page, users can select a town from a list of buttons. After choosing a town, they will see the number of resale transactions for each flat type in that area.



Flat Type	Transaction Count
4 ROOM	5146
5 ROOM	4339
3 ROOM	2659
EXECUTIVE	1457
2 ROOM	216

References

[1] Elmasri, R., & Navathe, S. B. (2016). *Fundamentals of Database Systems* (7th ed.). Pearson Education. A widely recognized text covering principles of database modeling, E/R diagrams, and relational schema mapping.

[2] Housing & Development Board (HDB). (n.d.). *Resale Flat Prices*. data.gov.sg. Retrieved from https://data.gov.sg/dataset/resale-flat-prices

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Processing math: 100%